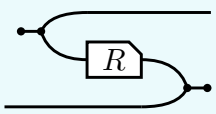
Relation Algebra

§3. ° : $\mathcal{C}^{op} \longrightarrow \mathcal{C}$ is a 2 functor

(3a)

° is primitive, part of the data the first section lists. It turns both of R's wires round, and the Frobenius structure DRAWS that — the picture and the formula below are R°, not its definition:



$$R^\circ = (\multimap\!\triangleleft \otimes \mathbb{1})(\mathbb{1} \otimes R \otimes \mathbb{1})(\mathbb{1} \otimes \triangleright\!\multimap)$$

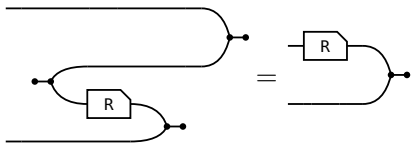
where $\multimap\!\triangleleft : \mathbb{I} \to a \otimes a$ opens a pair of wires out of nothing and $\triangleright\!\multimap : b \otimes b \to \mathbb{I}$ closes one, so the input of R° is where the output of R was. Taking that formula as the definition would now be circular: $\triangleleft$ and $\multimap$ are $\triangleright^\circ$ and $\multimap^\circ$. The laws of ° are that it is a contravariant 2-functor $^\circ : \mathcal{C}^{\circ p} \to \mathcal{C}$:

(i) $\mathbb{1}^\circ = \mathbb{1}$ (ii) $(RS)^\circ = S^\circ R^\circ$ (iii) $(R \otimes S)^\circ = R^\circ \otimes S^\circ$ (iv) $R \leq S$ implies $R^\circ \leq S^\circ$

§3.1. The slide

The one rule (ii) and (iii) use, and each of them uses it twice. A converse facing a merge on the lower strand is the box itself, upright, on the upper one:

(3.1a)



`R°` is DEFINED as the bending of `(R⊗𝟙)▷⊸`, so the slide claims only that unbending it gives that back — and unbending undoes bending for every arrow. That is the snake above with a passenger: the `⊸◁` bends the `a` strand down and the `▷⊸` brings it back up, while the `b` strand rides through untouched. Nothing else is spent below.